

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 25)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

- A Reactions that require high temperatures ☐
- B Reliance on non-renewable energy ☐
- C Waste products that are the raw materials for other processes ☐
- D Dispersal of elements so they cannot be re-used ☐

0 1 . 2

Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 1 . 3

Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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5

0 2 . 1

Explain the effects of deforestation on soil.

[4 marks]

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0 2 . 2

Explain how deforestation affects hydrology and climate.

[4 marks]

0 2 . 3

Give **two** ecological impacts of deforestation other than the direct loss of species.**[2 marks]**

1

2

10

0 3

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha ⁻¹ yr ⁻¹	Food energy output / GJ ha ⁻¹ yr ⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 3 . 1

Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

03 . 2

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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03 . 3

Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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03 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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10

04 . 1

Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

04 . 2

Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

04 . 3

Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

10

0 5 . 1

Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

0 5 . 2

Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

0 5 . 3

Describe **three** threats to UK broadleaf woodland.

[3 marks]

1

2

3

10

0 6 . 1

Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 6 . 2

Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

0 6 . 3

Explain how buying fairtrade food affects producers.

[1 mark]

0 6 . 4

Explain how nitrate from farmland can threaten human health.

[2 marks]

0 6 . 5

Give **two** agricultural sources of methane.

[2 marks]

1

2

10

0 7 . 1

Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

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Soil maintenance

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Interspecies relationships

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0 7 . 2

Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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07 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

10

08 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

08 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

08 . 2

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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08 . 3

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

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08 . 4

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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10

09 . 1

Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

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09 . 2

Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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09 . 3

Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

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09 . 4

Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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10

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

Extra space

10 . **2** State what is meant by a **fed** aquaculture species.

[1 mark]

10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

END OF QUESTIONS